

Lecture 10

①

Def Let $f \in L^1_{loc}(\mathbb{R}^n)$. For a measurable $E \subseteq \mathbb{R}^n$ with $|E| < \infty$

we write $\langle f \rangle_E = \frac{1}{|E|} \int_E f(x) dx$.

We say that f has bounded mean oscillation if

$$\|f\|_{BMO} := \sup_{\substack{Q \subseteq \mathbb{R}^n, \\ Q \text{ cube}}} \frac{1}{|Q|} \int_Q |f(x) - \langle f \rangle_Q| dx < \infty.$$

$= \langle |f - \langle f \rangle_Q| \rangle_Q$

If f, g have bounded mean oscillation

and $c \in \mathbb{C}$, then

$$\|f+g\|_{BMO} \leq \|f\|_{BMO} + \|g\|_{BMO} \quad \text{and}$$

$$\|cf\|_{BMO} = |c| \|f\|_{BMO}.$$

(2)

It follows that

$$\{ f \in L'_{loc}(\mathbb{R}^n) \mid \|f\|_{BMO} < \infty \}$$

is a linear space. However,

$\|\cdot\|_{BMO}$ is not a norm, since

$$\|f - g\|_{BMO} = 0 \text{ iff } f - g \text{ is}$$

constant a.e. If $\|f - g\|_{BMO} = 0$,

then we say that $f \sim g$. The

corresponding space of equivalence

classes

$$BMO(\mathbb{R}^n) := \{ [f]_{\sim} \mid f \in L'_{loc} \text{ and } \|f\|_{BMO} < \infty \}$$

is a complete linear space with

$$\text{norm } \|[f]_{\sim}\| = \|f\|_{BMO}.$$

③

Convention: We will disregard the equivalence structure of $BMO(\mathbb{R}^n)$, and instead say that $f \in BMO(\mathbb{R}^n)$ whenever $f \in L^1_{loc}$ and $\|f\|_{BMO} < \infty$.

Observation: Let $f \in L^1_{loc}$, and suppose for every bounded cube $Q \subseteq \mathbb{R}^n$ there exists $f_Q \in \mathbb{C}$ where $\{f_Q\}$ satisfies

$$A = \sup_Q \frac{1}{|Q|} \int_Q |f(x) - f_Q| dx < \infty.$$

Then $f \in BMO(\mathbb{R}^n)$. Indeed,

$$|f - \langle f \rangle_Q| \leq |f - f_Q| + |\langle f \rangle_Q - f_Q|.$$

$$\text{Since } |\langle f \rangle_Q - f_Q| = \left| \frac{1}{|Q|} \int_Q (f(x) - f_Q) dx \right| \leq A,$$

we may average the above inequality

(7)

to obtain that $\|f\|_{BMO} \leq 2A$.

This means that in the definition of $\|f\|_{BMO}$, we may replace $\langle f \rangle_Q$ with whatever other quantity that happens to be more convenient in the current situation.

Exercise: Prove that

$$\|f\|_{BMO} \approx \sup_{\substack{B \subset \mathbb{R}^n, \\ B \text{ ball}}} \frac{1}{|B|} \int_B |f(x) - \langle f \rangle_B| dx.$$

i.e. we may define $BMO(\mathbb{R}^n)$ using balls rather than cubes.

Exercise: Suppose $f \in L^\infty(\mathbb{R}^n)$. Then

$$\|f\|_{BMO} \leq 2\|f\|_\infty.$$

i.e. $L^\infty(\mathbb{R}^n) \subseteq BMO(\mathbb{R}^n)$.

(5)

Exercise: Let $f(x) = \log|x|$, $x \in \mathbb{R}^n$.

Then $f \in \text{BMO}(\mathbb{R}^n)$. i.e. $\text{BMO}(\mathbb{R}^n)$

also contains unbounded functions.

Exercise Let

$$f(x) = \begin{cases} \log x, & x > 0, \\ 0, & x \leq 0 \end{cases}$$

Then $f \notin \text{BMO}(\mathbb{R})$. i.e. $\text{BMO}(\mathbb{R}^n)$

is not stable under multiplication

with indicator functions. This is an

important difference between $\text{BMO}(\mathbb{R}^n)$

and $L^p(\mathbb{R}^n)$.

⑥

The examples show that BMO-functions may have logarithmic singularities. The main result of this lecture is that in a very precise sense this is the worst possible type of singularity for a BMO-function.

Theorem (John-Nirenberg)

There exists constants $C_1, C_2 \in (0, \infty)$ depending only on n such that for any $f \in \text{BMO}(\mathbb{R}^n)$, any cube Q , and any $\lambda > 0$

$$|\{x \in Q \mid |f(x) - \langle f \rangle_Q| > \lambda\}| \leq C_1 |Q| \cdot e^{-C_2 \lambda / \|f\|_{\text{BMO}}}$$

Remark: Allowing for values in $\overline{\mathbb{R}} = [-\infty, \infty]$, the above inequality makes sense also in

⑦

the degenerate cases $\|f\|_{BMO} = 0$ and

$$|Q| = \infty.$$

Before we prove the theorem, let's look at some consequences:

Prop Let $p \in (1, \infty)$. Then

$$\|f\|_{BMO} \approx \sup_Q \left(\frac{1}{|Q|} \int_Q |f(x) - \langle f \rangle_Q|^p dx \right)^{1/p}.$$

In particular, $BMO(\mathbb{R}^n) \subseteq \bigcap_{1 < p < \infty} L^p_{loc}(\mathbb{R}^n)$.

Pf By Jensen's inequality,

$$\left(\frac{1}{|Q|} \int_Q |f(x) - \langle f \rangle_Q| dx \right)^p \leq \frac{1}{|Q|} \int_Q |f(x) - \langle f \rangle_Q|^p dx.$$

As this holds for any Q , we obtain the inequality $LHS \leq RHS$.

⑧

For the reverse inequality, we compute

$$\begin{aligned} & \frac{1}{|Q|} \int_Q |f(x) - \langle f \rangle_Q|^p dx \\ &= \frac{1}{|Q|} \int_0^\infty p \lambda^{p-1} |\{x \in Q : |f(x) - \langle f \rangle_Q| > \lambda\}| d\lambda \end{aligned}$$

$$\leq C_1 \int_0^\infty p \lambda^{p-1} e^{-C_2 \lambda / \|f\|_{\text{BMO}}} d\lambda$$

John-Nirenberg.

$$\left[\frac{C_2 \cdot \lambda}{\|f\|_{\text{BMO}}} = s \right]$$

$$= C_1 \left(\frac{\|f\|_{\text{BMO}}}{C_2} \right)^p \int_0^\infty p s^{p-1} e^{-s} ds \leq \|f\|_{\text{BMO}}^p.$$

□

Remark: The proof reveals the power of the proposition; functions in $\text{BMO}(\mathbb{R}^n)$ are so non-singular that they satisfy a "reverse Jensen's inequality".

(9)

Another consequence: Let $f = \sum_{k=1}^{\infty} \lambda_k a_k$ be an atomic decomposition of $f \in \mathcal{H}'(\mathbb{R}^n)$, and $b \in \mathcal{BM}(\mathbb{R}^n)$. The linear functional $f \mapsto \langle f, b \rangle$ by

$$\langle f, b \rangle = \sum_{k=1}^{\infty} \lambda_k \int_{\mathbb{R}^n} a_k(x) b(x) dx.$$

Since each a_k has support on some bounded cube Q_k , and $a_k \in L^2(\mathbb{R}^n)$, the above integrals are well defined, because $b \in L^2_{loc}$.

If f is a finite linear combination of atoms then

$$\langle f, b \rangle = \int_{\mathbb{R}^n} f(x) b(x) dx,$$

which is the standard L^1 - L^∞ -duality pairing.

(10)

This might fail for general $f \in H'(\mathbb{R}^n)$,

since b might be unbounded.

However, $\langle f, b \rangle = \sum_{k=1}^{\infty} \tau_k \int_{\mathbb{R}^n} a_k(x) b(x) dx$

is still well-defined in the sense that

the series is absolutely convergent

and does not depend on the particular

decomposition of f . To see that the

series converges, note that

$$\left| \int_{\mathbb{R}^n} a_k(x) b(x) dx \right| = \left| \int_{Q_k} a_k(x) b(x) dx \right|$$

$$\text{supp } a_k \subseteq Q_k$$

$$\stackrel{\langle a_k \rangle_{Q_k} = 0}{=} \left| \int_{Q_k} a_k(x) (b(x) - \langle b \rangle_{Q_k}) dx \right|$$

$$\stackrel{\text{Cauchy-Schwarz}}{\leq} \|a_k\|_{L^2} \cdot \| (b - \langle b \rangle_{Q_k}) \chi_{Q_k} \|_{L^2} \leq \|b\|_{BMO}$$

$$\|a_k\|_{L^2}^2 \leq \frac{1}{|Q_k|} \text{ and } b \in BMO.$$

It follows that

$$|\langle f, b \rangle| \leq \|b\|_{\text{BMO}} \cdot \sum_{k=1}^{\infty} |\lambda_k|.$$

This proves not only that the series converges but that

$$|\langle f, b \rangle| \leq \|b\|_{\text{BMO}} \cdot \|f\|_{H^1},$$

i.e. the functional $\mathfrak{B}$ bounded.

This proves one half of the following important theorem:

Thm (C. Fefferman)

In the above sense, the dual space

$$H^1(\mathbb{R}^n)^* = \{ \text{bdd linear functionals on } H^1(\mathbb{R}^n) \}$$

coincides with $\text{BMO}(\mathbb{R}^n)$.

(2)

Remark: Fefferman's result was originally proved using different methods. The result later inspired the characterization of $H^1(\mathbb{R}^n)$ using atoms:

The atomic definition of $H^1(\mathbb{R}^n)$ is one where Fefferman's theorem is particularly easy to prove!

Consequence for CZOs:

Let T be a CZO of convolution type. When we discussed their boundedness on L^p , we first considered $p \in (1, 2]$. We then used certain symmetry properties to say that if T is bounded on L^p , then

(13)

it's also bounded on the dual space $(L^p)^*$, which happens to be $L^{p'}$, where $\frac{1}{p} + \frac{1}{p'} = 1$. By a similar argument, one can use the boundedness of $T: H^1(\mathbb{R}^n) \rightarrow L^1(\mathbb{R}^n)$ to infer boundedness of $T: (L^1(\mathbb{R}^n))^* \rightarrow (H^1(\mathbb{R}^n))^*$.

By Fefferman's theorem, we therefore obtain:

Theorem Let T be a CZO of convolution type. Then

$$T: L^\infty(\mathbb{R}^n) \rightarrow BMO(\mathbb{R}^n)$$

is bounded.

(14)

Pf of the John-Nirenberg thm

The inequality

$$|\{x \in Q \mid |f(x) - \langle f \rangle_Q| > \lambda\}| \leq C_1 |Q| \cdot e^{-C_2 \lambda / \|f\|_{\text{BMO}}}$$

does not change if we replace

f with cf , $c \in \mathbb{C}$. (We just

replace one arbitrary $\lambda > 0$ with

another.) Hence, we may assume

$$\text{that } \|f\|_{\text{BMO}} = 1.$$

We first note that if λ is

small, say $\lambda \in (0, 2^{n+1})$, then

$$|\{x \in Q \mid |f(x) - \langle f \rangle_Q| > \lambda\}| \leq |Q| \leq |Q| \cdot e^{C_2(2^{n+1} - \lambda)}$$

For an arbitrary choice of $C_2 > 0$.

For $\lambda \in (0, 2^{n+1})$ we therefore

have that

$$|\{x \mid |f(x) - \langle f \rangle_Q| > \lambda\}| \leq e^{C_2} \cdot |Q| \cdot e^{-C_2 \lambda},$$

where $C_2 > 0$ can be anything.

Suppose now instead that

$$\lambda \in [N \cdot 2^{n+1}, (N+1) \cdot 2^{n+1}).$$

The idea of the proof is to repeat the argument from the Calderon-Zygmund decomposition N times; with smaller starting cubes in each step. This

will eventually produce a family

$\{Q_{j_1, j_2, \dots, j_N}\}$ of cubes with the properties that

$$|f(x) - \langle f \rangle_Q| \leq N \cdot 2^{n+1} \text{ whenever } x \in Q \setminus \bigcup_{j_1, \dots, j_N} Q_{j_1, \dots, j_N}$$

(16)

and

$$\sum |\mathcal{Q}_{j_1, \dots, j_N}| < \frac{|\mathcal{Q}|}{2^N}.$$

It then follows that

$$|\{x \in \mathcal{Q} \mid |f(x) - \langle f \rangle_{\mathcal{Q}}| > \lambda\}|$$

$$\leq |\{x \in \mathcal{Q} \mid |f(x) - \langle f \rangle_{\mathcal{Q}}| > N \cdot 2^{n+1}\}|$$

$$\leq |\cup \mathcal{Q}_{j_1, \dots, j_N}| \leq \frac{|\mathcal{Q}|}{2^N}.$$

To obtain the inequality we

want, we just need to control 2^{-N} by $C_1 \cdot e^{-C_2 \cdot \lambda}$:

$$\begin{aligned} 2^{-N} &= 2 \cdot 2^{-(1+N)} \leq 2 \cdot 2^{-\frac{\lambda}{2^{n+1}}} = 2 \cdot \left(2^{\frac{1}{2^{n+1}}}\right)^{-\lambda} \\ &= 2 \cdot e^{-\frac{\log 2}{2^{n+1}} \cdot \lambda}. \end{aligned}$$

It now remains to explain the iterated Calderon-Zygmund decomposition:

In the first step, let

$$f_0(x) = (f(x) - \langle f \rangle_Q) \chi_Q(x).$$

Then

$$\frac{1}{|Q|} \int_Q |f_0(x)| dx \leq \|f\|_{BMO} = 1.$$

Bisect the sides of Q to obtain 2^n "children" of Q .

Among these children, select any cubes Q_j for which

$$\frac{1}{|Q_j|} \int_{Q_j} |f_0(x)| dx > 2.$$

Since $\frac{|Q|}{|Q_j|} = 2^n$, it also holds

that

$$\frac{1}{|Q_j|} \int_{Q_j} |f_0(x)| dx \leq 2^{n+1}.$$

Repeat this process of bisection and selection on any cube not already selected. This gives a collection of cubes $\{Q_{j_i}\}$ with disjoint interiors such that

$$2 < \frac{1}{|Q_{j_i}|} \int_{Q_{j_i}} |f_0(x)| dx \leq 2^{n+1},$$

$$|f_0(x)| \leq 2 \text{ for a.e. } x \in Q \setminus \bigcup Q_{j_i}.$$

It follows that

$$\sum |Q_{j_i}| < \sum \frac{1}{2} \int_{Q_{j_i}} |f_0(x)| dx$$

$$= \frac{1}{2} \int_{\bigcup Q_{j_i}} |f_0(x)| dx \leq \frac{1}{2} \int_Q |f_0(x)| dx$$

$$\leq \frac{1}{2} |Q|.$$

(19)

This gives us the desired collection of cubes $\{Q_{j_1, \dots, j_N}\}$ when $N=1$.

Assume now that we have a collection of k cubes $\{Q_{j_1, \dots, j_N}\}$ with disjoint interiors, and such that

$$\sum |Q_{j_1, \dots, j_N}| \leq \frac{|Q|}{2^N}$$

and

$$|f(x) - \langle f \rangle_Q| \leq N \cdot 2^{n+1} \text{ for a.e.}$$

$$x \in Q \setminus \bigcup Q_{j_1, \dots, j_N}.$$

Moreover,

$$2 < \frac{1}{|Q_{j_1, \dots, j_N}|} \int_{Q_j} |f(x) - \langle f \rangle_{Q_{j_1, \dots, j_{N-1}}}| dx \leq 2^{n+1}.$$

(20)

For each cube $Q_{j_1, \dots, j_N}$, define

$$f_{j_1, \dots, j_N}(x) = (f(x) - \langle f \rangle_{Q_{j_1, \dots, j_N}}) \chi_{Q_{j_1, \dots, j_N}}(x).$$

Construct a collection of

sub cubes $\{Q_{j_1, \dots, j_{N+1}}\}$ s.t.

$$\sum |Q_{j_1, \dots, j_{N+1}}| \leq \frac{|Q_{j_1, \dots, j_N}|}{2}$$

$$2 < \frac{1}{|Q_{j_1, \dots, j_{N+1}}|} \int_{Q_{j_1, \dots, j_{N+1}}} |f_{j_1, \dots, j_N}(x)| dx \leq 2^{n+1},$$

$$|f_{j_1, \dots, j_N}(x)| \leq 2 \quad \text{for a.e. } x \in$$

$$Q \cup Q_{j_1, \dots, j_{N+1}}.$$

(21)

FINAL

It follows that for a.e.

$$x \in Q \cup Q_{j_1, \dots, j_{N+1}},$$

$$|f(x) - \langle f \rangle_Q| \leq |f(x) - \langle f \rangle_{Q_{j_1, \dots, j_N}}|$$

$$+ |\langle f \rangle_{Q_{j_1, \dots, j_N}} - \langle f \rangle_{Q_{j_1, \dots, j_{N-1}}}|$$

$$+ \dots + |\langle f \rangle_{Q_{j_1}} - \langle f \rangle_Q| \leq (N+1) \cdot 2^{N+1}.$$

□